Indian Institute of Technology Jodhpur

Discrete Mathematics (MAL2190)

Project Report

Build a Game, Discover the Mathematics

[Title of the game or puzzle]

Group number	[Group number]
Submission date	[DD Month YYYY]
Concept(s)	[Main discrete mathematics concept(s)]

Group members (10 students)

No.	Name	Roll number
1	*[Name of member 1]*	*[Roll number]*
2	*[Name of member 2]*	*[Roll number]*
3	*[Name of member 3]*	*[Roll number]*
4	*[Name of member 4]*	*[Roll number]*
5	*[Name of member 5]*	*[Roll number]*
6	*[Name of member 6]*	*[Roll number]*
7	*[Name of member 7]*	*[Roll number]*
8	*[Name of member 8]*	*[Roll number]*
9	*[Name of member 9]*	*[Roll number]*
10	*[Name of member 10]*	*[Roll number]*

Using this template. Replace all bracketed text with your group's work. Keep the main section headings and formatting so the reports can be combined for the course website. Add space and pages where needed; the preview length is not a prescribed page limit. Remove guidance boxes before submission. Explain the game so that a reader can recreate and play it.

1. The Game or Puzzle

1.1 Overview and objective

[Introduce the game in a short paragraph. State what the players are trying to achieve and what makes the activity interesting.]

1.2 Players, materials, and initial setup

- **Number of players:** *[Individual, two-player, team, or other format.]*
- **Materials or platform:** *[List the board, cards, counters, software, or other items needed. For a digital game, explain how to access and start it.]*
- **Initial setup:** *[Describe the starting arrangement and how the first player or first move is chosen, if applicable.]*

1.3 Rules and how to play

Write the rules in the order a new player needs them. State the legal moves, restrictions, turn order, and ending condition. Define winning, losing, scoring, ties, or puzzle completion wherever relevant.

1. *[First rule or step.]*
2. *[Second rule or step.]*
3. *[Further rules and restrictions; add items as needed.]*
4. *[When the game ends and how the outcome is decided.]*

1.4 Illustrative play-through

[Show one short example from a valid starting state through the main moves to an outcome. Use a labelled diagram, photograph, or table if it helps. Explain why each move is allowed.]

2. The Mathematical Question

[State the precise mathematical question that arises from the game. What do you want to determine, count, characterise, or prove?]

Connection to discrete mathematics. *[Name the relevant concept(s) and explain how the game represents them. Define the symbols, objects, and assumptions used in your analysis.]*

3. Experimentation and What We Discovered

3.1 Method and observations

[Explain how you explored the game: which starting positions, sizes, strategies, or rule variations you tested, and how you recorded the outcomes. If you used code, describe what it checked.]

Record actual trials. Distinguish one play-through from an exhaustive check of all possibilities. Add or modify rows and columns to suit your game.

Trial	Starting case	Moves or strategy tested	Outcome / observation
1	*[Case]*	*[What you tried]*	*[Result]*
2	*[Case]*	*[What you tried]*	*[Result]*
3	*[Case]*	*[What you tried]*	*[Result]*

3.2 Patterns and discoveries

[Describe the patterns you noticed and the evidence for them. Mention any surprising outcomes, exceptions, or unsuccessful strategies. Explain how these observations led to your conjecture.]

4. Our Strategy, Conjecture, or Mathematical Claim

4.1 Precise statement

State the range of cases and all conditions under which your claim is meant to hold. For a winning strategy, specify which player can use it, from which positions, and whether it must work against every legal opponent move. Repeated wins alone do not establish a guaranteed winning strategy.

Claim. *[Write your conjecture or mathematical claim as a complete, precise statement.]*

4.2 Strategy and intuition

[Describe the proposed strategy step by step, if applicable. Explain the intuition behind it and how it follows from your observations. If the claim is about counting, structure, or impossibility, explain the main idea instead.]

5. Proof or Counterexample

Complete the appropriate route below and remove the unused route. Experiments may suggest a claim; a proof must establish it for every case covered by the statement. A valid counterexample must satisfy the assumptions and contradict the claimed conclusion. Support your main claim with a proof or disprove it with a counterexample.

Route A: Proof of the claim

Proof. *[Give a clear, complete argument. State any facts or lemmas used and justify the important steps. Use a suitable method, such as induction, contradiction, a direct argument, an invariant, a counting argument, or a graph argument. Choose the method that actually fits your claim.]*

[For induction, show the base case(s), state the induction hypothesis, and prove the induction step. For a winning strategy, explain why each prescribed move is legal, why every possible opponent response is covered, and why the desired outcome is eventually reached. Remove this prompt if it does not apply.]

Conclusion of the proof. *[State exactly what has been established and for which cases.]*

Route B: Counterexample to the conjecture

Original conjecture. *[Restate the conjecture being tested.]*

Counterexample. *[Give the specific starting state, construction, or sequence of legal moves. Explain why it satisfies the assumptions and why the conclusion fails. A loss caused by ignoring the proposed strategy is not a counterexample to that strategy.]*

What the counterexample teaches us. *[Explain which part of the conjecture fails. If you propose a revised claim, state it precisely and distinguish a proved result from a conjecture that remains unproved.]*

Final mathematical finding

[Summarise what you can now conclude about the game and whether your main claim was proved or disproved. State the cases covered by your result. Identify any further unanswered questions separately.]

6. Interaction with Students and Feedback

The participants may be up to 18 years old. If the activity has not yet taken place, label this section **Planned interaction**; describe the plan and leave actual observations for later. Do not present planned outcomes as observed results.

Introduction and guided play. *[Explain how you will introduce the rules simply, demonstrate a round, and help students play. Give two or three guiding questions or hints that encourage them to discover the mathematical idea.]*

Mathematics after playing. *[Write a brief, age-appropriate explanation of the idea you will discuss after the students have explored the game.]*

Feedback and refinements, if available. *[Record what students or classmates found clear, confusing, or interesting. Explain any changes made to the rules, materials, explanation, or strategy. If no testing or interaction has occurred, state that clearly.]*

7. Contribution of Each Group Member

Use the same member order as on the first page. Describe each person's specific contribution; shared tasks may appear in more than one row. Examples include game design, construction, testing, mathematical analysis, proof, illustrations, writing, and facilitating the activity.

No.	Name / roll number	Specific contribution(s)
1	*[Member 1]*	*[Contribution]*
2	*[Member 2]*	*[Contribution]*
3	*[Member 3]*	*[Contribution]*
4	*[Member 4]*	*[Contribution]*
5	*[Member 5]*	*[Contribution]*
6	*[Member 6]*	*[Contribution]*
7	*[Member 7]*	*[Contribution]*
8	*[Member 8]*	*[Contribution]*
9	*[Member 9]*	*[Contribution]*
10	*[Member 10]*	*[Contribution]*

8. References and Acknowledgements

[List any books, websites, games, software, or other resources you used. If you adapted an existing game or proof, credit the source and explain your adaptation. If no external sources were used, state this.]